

Encoding and Calibration of Exact Norm Penalties in Mixed-Integer Convex Optimization

Abstract

Finite exactness does not ensure that a norm penalty has a short binary representation or that its smallest sufficient coefficient can be approximated efficiently. Our main contribution is to quantify this gap for mixed-integer convex quadratic models. A compact family with one binary variable, $d + 1$ continuous variables and fixed rational coefficients has least penalty 2^{2^d-1} , even after optimizing an unrestricted scalar linking multiplier. Thus every sufficient rational coefficient needs at least 2^d numerator bits. This obstruction persists at fixed additive accuracy and with uniform strict points on feasible integer slices. Conversely, for quadratic input of length N with n continuous variables, explicit finite boxes and refined Slater on every equality-feasible slice, there is a sufficient integer zero-multiplier coefficient with $(N + 1)2^{O(n)}$ bits for the infinity and one-norm penalties, preserving the full minimizer set. A second bound is $N^{O(k+1)}$ bits when only k native inequalities are nonlinear in the continuous variables, allowing arbitrarily many affine rows. Two complementary results distinguish representation from calibration and conditioning. Calibration within any polynomial factor of the least optimized-dual penalty is NP-hard even on a binary box with a known unique primal optimizer and an obvious sufficient coefficient. Finite rational right-hand-side perturbations instead give short sufficient infinity-norm coefficients with high probability for finite unions of compact convex slices with continuous convex objectives and affine residuals. This guarantee concerns the perturbed problem and allows infeasibility; examples delimit its dependence on the slice count and its implications for numerical size and expectations.

1 Introduction

An exact norm penalty replaces a linking equality system by a finite charge for its violation without changing the optimal value. Existence alone does not say whether that coefficient can be stored or produced efficiently. Nor does an easily available sufficient coefficient make the smallest one easy to find. We study these two questions using ordinary binary encoding: *representation* asks for the number of bits in a sufficient coefficient, whereas *calibration* asks how closely one can approximate the least sufficient coefficient.

The main contribution is a quantitative account of what changes when convex quadratic constraints are added to the native mixed-integer set. Here *native* refers to all constraints retained while a separate affine equality system is penalized. Our lower bound applies even after maximizing over all linking multipliers, so it cannot be removed by a better multiplier choice. Our upper bounds give equality of the full minimizer sets at multiplier zero, a stronger conclusion than equality of optimized dual values. Section 2 states these distinctions precisely.

Main results and their significance. Theorem 3.1 gives a compact convex-quadratic family with chain length d , one binary variable, a linear objective and $d + 1$ continuous variables whose least optimized-dual penalty is 2^{2^d-1} . The data use a fixed finite set of rational coefficients and

$O(d \log(d + 1))$ bits, but every exact rational penalty needs at least 2^d numerator bits. Feasible integer slices have uniformly strict points, and fixed additive accuracy below the unit objective range does not remove the obstruction (Corollary 3.2). Thus finite norm-penalty exactness for convex quadratic constraints does not extend the polynomial encoding guarantee known for linear native constraints. The obstruction concerns the penalty representation, not the difficulty of finding the original optimum, which is immediate in this family.

Theorems 4.1 and 5.1 supply positive bounds under explicit finite boxes and refined Slater on every equality-feasible slice. For input length N and n continuous variables, there exists a sufficient integer coefficient with $(N + 1)2^{O(n)}$ bits for both the infinity and one-norm penalties. A second bound is $N^{O(k+1)}$ bits if at most k native inequalities are nonlinear in the continuous variables, with no restriction on the number of affine rows or integer assignments. Within this model, these statements locate the obstruction in the growing nonlinear continuous structure: fixing n or k permits polynomial-time output of a conservative coefficient (Corollary 5.3). They do not provide a practical small coefficient or an algorithm for solving the penalized subproblem.

The complementary results separate two further issues. Theorem 6.3 rules out polynomial-time calibration within any fixed polynomial factor, unless $P = NP$, on a binary box with one equality, a linear objective, a known unique primal optimizer and a known sufficient coefficient of one. Both the optimized-multiplier and zero-multiplier thresholds have this hardness. Section 7 gives a different route to short coefficients: a sampled rational right-hand side, with probability at least $1 - \epsilon$, is either infeasible or admits the explicit zero-multiplier infinity-norm coefficient $4mKM/(\sigma\epsilon)$. The slices are compact and convex, with continuous convex objectives and affine residuals. Here m is the residual dimension, K bounds the number of slices, M is their common objective range, and σ is the noise-cube radius. The finite-grid guarantee includes the sampling precision and coefficient encoding. It applies to the sampled problem; it does not guarantee preservation of the original optimizer or bound the expected numerical penalty.

Relation to prior work. Boland and Eberhard [9] establish finite-penalty exactness for bounded pure integer programs under their augmentation assumptions; Feizollahi et al. [12] establish norm-penalty exactness for rational MILPs. For a convex quadratic objective over a rational mixed-integer polyhedron, Gu et al. [15, Theorem 11] prove polynomial binary encoding of an exact norm penalty. Bhardwaj et al. [7, Theorems 3.1–3.3 of the inspected preprint] give value-function proofs of existence and data-dependent penalty parameters for MILPs and MIQPs. For nonlinear convex objectives over polyhedral mixed-integer sets they also prove exactness, with an explicit bound under strong convexity and smoothness. For nonlinear native constraints, Lefebvre and Schmidt [19, Theorem 14] prove finite exactness at each fixed multiplier under compactness, differentiable convex objectives, convex continuous slices and refined Slater on every equality-feasible slice. Their December 15, 2025 manuscript asks whether polynomial penalty encoding extends to quadratically constrained models and whether a smallest exact penalty can be computed in polynomial time. Our lower bound answers the first question negatively in growing continuous dimension; the binary-box reduction answers the second negatively in growing binary dimension, subject to $P \neq NP$.

The lower-bound mechanism has important precedents. Bienstock et al. [8, Section 6 of the inspected preprint] give quadratic large-coordinate and small-violation examples; their nonconvex Example 6.2 already implies large zero-multiplier norm penalties. Beck et al. [6, Sections 2 and 4, Result 4] use a bounded convex squaring chain with Slater regularity to show sensitivity of bilevel problems to feasibility errors. We do not claim the squaring chain or its small residuals as new. To the best of our knowledge, the new lower-bound statement is the exact optimized-dual threshold in Theorem 3.1 for a compact convex-quadratic model with one binary variable and uniform strict

points, together with its fixed-accuracy consequence. Opposite-sign residuals are what make the bound hold for every multiplier.

The upper bounds use established real-algebraic tools: component-radius bounds [4] and quantitative elimination [5]. Few-quadratic methods already control algebraic complexity in the number of quadratic constraints [14, 18]. The contribution here is the resulting penalty encoding bounds, including the dependence on the number of *nonlinear native* inequalities while allowing arbitrarily many affine rows. To the best of our knowledge, these bounds for the boxed convex-slice model of Section 4 are not established in the cited exact-penalty or few-quadratic results. The linear-constraint case and polynomial-time conservative output in that case are prior results, not new consequences claimed here; Section 5 gives the detailed comparison.

Generic penalty-choice hardness, even with a known constrained optimum, also predates this paper [1]. The distinction in Theorem 6.3 is the polynomial-factor conclusion for an always-positive norm-penalty threshold after unrestricted multiplier optimization, under the binary-box restrictions above. We are not aware of that restricted conclusion in the prior tuning results compared in Section 6. Finally, distance to convex boundaries is a classical conditioning device [10, 11]. Our perturbation result is an explicit application to exact penalties on finitely many convex slices, with rational-grid and feasibility qualifications; we make no separate priority claim for the boundary-avoidance principle, the Gaussian tube estimate or the convex repair argument.

2 Exactness and residual geometry

Let X be a nonempty compact subset of a finite-dimensional Euclidean space. Let $f : X \rightarrow \mathbb{R}$ and $r : X \rightarrow \mathbb{R}^m$ be continuous, and suppose $F = \{x \in X : r(x) = 0\}$ is nonempty. Fix a norm on $\mathbb{R}^m$ and write

$$v = \min_{x \in F} f(x), \quad L_\rho(\lambda) = \min_{x \in X} \{f(x) + \lambda^T r(x) + \rho \|r(x)\|\}, \quad D_\rho = \sup_{\lambda \in \mathbb{R}^m} L_\rho(\lambda), \quad (1)$$

where $\rho \geq 0$. The set X retains the *native constraints*; only $r(x) = 0$ is relaxed. Weak duality gives $L_\rho(\lambda) \leq D_\rho \leq v$. We call ρ *dual-value exact* if $D_\rho = v$, and exact at a specified multiplier λ if $L_\rho(\lambda) = v$. The latter includes attainment of the dual supremum. We call (ρ, λ) *minimizer-set exact* if the minimizers in $L_\rho(\lambda)$ are exactly $\operatorname{argmin}_F f$. The distinctions are necessary even on compact sets.

Set $\rho_* = \inf\{\rho \geq 0 : D_\rho = v\}$, with $\inf \emptyset = +\infty$. For a fixed multiplier, the corresponding value threshold is

$$\rho_*(\lambda) = \max \left\{ 0, \sup_{x \in X \setminus F} \frac{v - f(x) - \lambda^T r(x)}{\|r(x)\|} \right\}. \quad (2)$$

An empty supremum contributes no positive lower bound. This formula follows by requiring every penalized objective value to be at least v . A finite threshold is exact at that multiplier. Every strictly larger coefficient is minimizer-set exact: its additional penalty is positive at each infeasible point. In contrast, finite ρ_* need not give an attaining multiplier.

Primal convexification descriptions of augmented Lagrangian duals are established [9, 12]. In particular, Feizollahi et al. [12, Theorem 1 of the inspected manuscript] use a lifted convex hull under a relative-interior feasibility condition. The following residual-space analogue for compact X needs no relative-interior condition. Its proof does not assume multiplier attainment; we make no novelty claim for the convexification argument.

Proposition 2.1 (Balanced mixtures). *For every $\rho \geq 0$,*

$$D_\rho = \min \left\{ \sum_{i=1}^p \theta_i [f(x_i) + \rho \|r(x_i)\|] : \begin{array}{l} 1 \leq p \leq m+1, \ x_i \in X, \ \theta_i \geq 0, \\ \sum_i \theta_i = 1, \ \sum_i \theta_i r(x_i) = 0 \end{array} \right\}. \quad (3)$$

Proof. The convex hull K of $\{(r(x), f(x) + \rho \|r(x)\|) : x \in X\}$ is compact. Let $w = \min\{t : (0, t) \in K\}$. Averaging over a balanced mixture cancels λ , so $D_\rho \leq w$. For $a < w$, strict separation of $(0, a)$ from K gives $u^T z + \beta t > \beta a$ for every $(z, t) \in K$. At $(0, w)$ this implies $\beta > 0$. Compactness then gives $L_\rho(u/\beta) > a$. Letting $a \uparrow w$ proves $D_\rho = w$, without assuming multiplier attainment.

Carathéodory's theorem gives an optimal mixture with at most $m+2$ points. If it has more than $m+1$ positive weights, the columns $(1, r(x_i))$ are linearly dependent. Both signs of a sufficiently small weight perturbation along a dependence preserve feasibility. Optimality forces its objective derivative to be zero. Increase the perturbation until a weight vanishes; the value is unchanged. This reduces the support to at most $m+1$. $\square$

The average in (3) contains the norms of the individual residuals, not the norm of their average, which is zero. The support bound is sharp: take one point with $(r, f) = (0, 0)$ and $m+1$ points with cost -1 and residuals $e_1, \dots, e_m, -\sum_j e_j$. At $\rho = 0$, an optimal balanced mixture must give weight $1/(m+1)$ to each of the latter points.

Corollary 2.2. *Let $\mathcal{B}$ denote the balanced mixtures in (3), and write $c_\mu = \sum_i \theta_i f(x_i)$ and $s_\mu = \sum_i \theta_i \|r(x_i)\|$. Then*

$$\rho_* = \max \left\{ 0, \sup_{\mu \in \mathcal{B}: s_\mu > 0} \frac{v - c_\mu}{s_\mu} \right\}. \quad (4)$$

If this number is finite, it is dual-value exact.

Proof. A mixture with $s_\mu = 0$ uses only feasible points of positive weight, so $c_\mu \geq v$. The remaining inequalities in (3) are $\rho s_\mu \geq v - c_\mu$. They all hold at their finite supremum. $\square$

Proposition 2.3 (One relaxed equality). *Suppose $m = 1$, use absolute value, and assume residuals of both signs occur. Define*

$$A_+ = \sup_{r(x) > 0} \frac{v - f(x)}{r(x)}, \quad A_- = \sup_{r(x) < 0} \frac{v - f(x)}{-r(x)}.$$

If both are finite, then

$$\rho_* = \max\{0, (A_+ + A_-)/2\}, \quad L_\rho(\lambda) = v \iff A_+ - \rho \leq \lambda \leq \rho - A_-. \quad (5)$$

If either is infinite, no finite penalty is dual-value exact. At multiplier zero the threshold is $\max\{0, A_+, A_-\}$.

Proof. The pointwise inequalities for positive and negative residuals give the interval in (5). To see that an exact supremum cannot avoid this interval, fix one point of each residual sign. Evaluating any sequence $L_\rho(\lambda_j) \rightarrow v$ at these two points bounds λ_j above and below. Along a convergent subsequence, upper semicontinuity of L_ρ gives an attaining multiplier. Hence exactness requires and is implied by a nonempty interval, giving all the claims. $\square$

If residuals occur on only one side of zero, every balanced mixture is supported on F , so $D_\rho = v$ for all $\rho \geq 0$. Attainment can fail: for $X = [0, 1]$, $r(x) = x$, and $f(x) = -\sqrt{x}$, every finite multiplier and penalty give $L_\rho(\lambda) < 0 = v$. Appendix A records the same distinction in a mixed-integer quadratic example from the literature.

Lemma 2.4 (A bound from integer slices). *Suppose X is the union of finitely many nonempty compact slices X_z , $f \geq L$ on X , and $v \leq U$. On every slice meeting F , suppose there is λ_z such that*

$$f(x) + \lambda_z^T r(x) \geq v_z := \min_{X_z \cap F} f, \quad \|\lambda_z\|_* \leq M \quad (x \in X_z),$$

where $\|\cdot\|_*$ is the dual norm. Suppose every other slice has $\min_{X_z} \|r\| \geq \delta > 0$. Then

$$\rho \geq \max\{M, (U - L)/\delta\} \quad (6)$$

is exact at multiplier zero. Any strictly larger coefficient is minimizer-set exact. If all slices meet F , omit the second term.

Proof. On a feasible slice, the dual-norm inequality gives $f + \rho\|r\| \geq f + \lambda_z^T r \geq v_z \geq v$. On an infeasible slice, $f + \rho\|r\| \geq L + \rho\delta \geq U \geq v$. A primal minimizer attains v . Strictly increasing ρ excludes every infeasible point from the minimizing set. $\square$

Thus feasible-slice multiplier bounds and infeasible-slice separation control different parts of the penalty. For convex slices, Slater regularity can supply the former without giving a useful quantitative bound on the latter.

3 An exponential encoding lower bound

All encoding statements use explicit rational coefficients and ordinary binary representations of their numerators and positive denominators. A sparse encoding records variable indices as well as nonzero coefficients. The norm and the relaxed equality are fixed as displayed.

For $n \geq 1$, let

$$A_n = \{a \in [0, 1]^n : a_1 \geq 1/4, a_{i+1} \geq a_i^2 \ (1 \leq i < n)\}, \quad \delta_n = 2^{-2^n}. \quad (7)$$

Induction shows $a_i \geq 2^{-2^i}$, with all bounds attained by $\bar{a}_i = 2^{-2^i}$. Define

$$X_n = \{(a, y, q) : a \in A_n, -1 \leq y \leq 1, q \in \{0, 1\}, y \geq a_n - 2(1 - q)\}. \quad (8)$$

We minimize $-q$ subject to $(a, y, q) \in X_n$ and $y = 0$. In (1), take $f = -q$, $r = y$, and $\|r\| = |y|$.

Theorem 3.1. *The problem defined using X_n in (8) has value $v_n = 0$, and for every $\rho \geq 0$,*

$$D_{n,\rho} = \min \left\{ 0, \frac{2\rho\delta_n - 1}{1 + \delta_n} \right\}. \quad (9)$$

Its dual supremum is attained and its least dual-value exact penalty is

$$\rho_{*,n} = \frac{1}{2\delta_n} = 2^{2^n-1}. \quad (10)$$

The instance has one binary variable, $n + 1$ continuous variables, a linear objective, and convex quadratic native constraints with coefficients from a fixed finite set. Its sparse encoding length is $O(n \log(n + 1))$, whereas every rational exact penalty needs at least 2^n numerator bits. Every equality-feasible integer slice and every native integer slice have strict points whose continuous inequality slacks are at least $1/8$.

Proof. Projection onto (q, y) gives exactly

$$(\{0\} \times [-1, 1]) \cup (\{1\} \times [\delta_n, 1]). \quad (11)$$

The inclusion follows from the chain; the reverse follows by setting $a = \bar{a}$. At $q = 0$ the lower bound $\delta_n - 2$ is below -1 . Consequently $y = 0$ forces $q = 0$ and $v_n = 0$.

For any multiplier, the points $(q, y) = (0, -1)$ and $(1, \delta_n)$ give values $\rho - \lambda$ and $-1 + (\rho + \lambda)\delta_n$. Their weighted average, with weights $\delta_n/(1 + \delta_n)$ and $1/(1 + \delta_n)$, is $(2\rho\delta_n - 1)/(1 + \delta_n)$. The point $(0, 0)$ gives zero. These witnesses prove the upper bound in (9) for every multiplier.

For $0 \leq \rho \leq 1/(2\delta_n)$, take

$$\lambda_\rho = \frac{1 + \rho(1 - \delta_n)}{1 + \delta_n} \geq \rho.$$

On the first interval of (11), the minimum occurs at $y = -1$; on the second it occurs at $y = \delta_n$. Their values coincide with the proposed dual value. For $\rho \geq 1/(2\delta_n)$, take $\lambda = \rho$. The penalized objective is nonnegative on both intervals and zero at $(0, 0)$. This proves the formula and attainment.

Each inequality $a_i^2 - a_{i+1} \leq 0$ is convex; all other constraints are affine. Across all constraints there are $O(n)$ nonzero coefficients, each with variable indices of $O(\log(n + 1))$ bits. The integer $2^{2^n - 1}$ has exactly 2^n bits. If a rational p/d with positive integers p, d is at least this value, then $p \geq 2^{2^n - 1}$ as well.

For the equality-feasible slice $q = 0$, set $a_i = 3/8$ and $y = 0$. For the native slice $q = 1$, set $a_i = 3/8$ and $y = 1/2$. The seed inequality has slack $1/8$, each chain inequality has slack $15/64$, and the variable bounds and linking inequality have slack at least $1/8$. The native $q = 0$ slice uses the first point. These are uniform strict points, although the equality-infeasible $q = 1$ slice comes within δ_n of $y = 0$. $\square$

The construction satisfies Assumptions 1–5 of Lefebvre and Schmidt [19], so their Theorem 14 ensures a finite exact penalty at every fixed multiplier. Theorem 3.1 concerns the encoding length of such penalties, even after optimizing the multiplier.

At $\rho = \rho_{*,n}$, Proposition 2.3 gives the unique exact multiplier $\lambda = \rho_{*,n}$: here $A_+ = 1/\delta_n$ and $A_- = 0$. Both witnesses in the proof then tie with feasible optima. For every $\rho > \rho_{*,n}$, the fixed multiplier $\lambda = \rho_{*,n}$ makes all infeasible objective values positive, so the minimizer set is exact. The zero-multiplier value threshold is $1/\delta_n$, twice the optimized threshold.

Corollary 3.2 (Fixed additive accuracy). *For $0 \leq \epsilon < 1$, the condition $v_n - D_{n,\rho} \leq \epsilon$ is equivalent to*

$$\rho \geq \max \left\{ 0, \frac{1 - \epsilon(1 + \delta_n)}{2\delta_n} \right\}. \quad (12)$$

For each fixed $\epsilon < 1$ and all sufficiently large n , such a rational penalty needs at least $2^n - O_\epsilon(1)$ numerator bits.

Proof. The first claim follows from (9). For large n , $\epsilon\delta_n \leq (1 - \epsilon)/2$, so the right-hand side is at least $(1 - \epsilon)/(4\delta_n)$. Taking binary logarithms gives the bit bound. $\square$

In particular, accuracy $\epsilon = 1/2$ requires $\rho \geq (1 - \delta_n)/(4\delta_n)$. The objective range is one, so this is a fixed accuracy relative to that normalization. The lower bound is exponential in continuous dimension and superpolynomial in input length; it is not a claim of an exponential bound in the full sparse input length. Nor is it a running-time lower bound for solving the primal, whose value is immediate from (11).

3.1 A symmetric variant and the role of the penalty function

A two-binary companion makes multiplier cancellation symmetric. Replace the final inequality in (8) by

$$y \geq a_n - (1 - q) - 2(1 - b), \quad y \leq -a_n + (1 - q) + 2b, \quad b \in \{0, 1\}. \quad (13)$$

Keep the same objective and relaxed equality. Projection now gives

$$(\{0\} \times [-1, 1]) \cup (\{1\} \times ([-1, -\delta_n] \cup [\delta_n, 1])).$$

Indeed, at $a = \bar{a}$ the two $q = 0$ intervals are $[-1, 1 - \delta_n]$ and $[\delta_n - 1, 1]$, which cover $[-1, 1]$; at $q = 1$ they are the two outer intervals. If $c = \rho - |\lambda|$, minimizing on these intervals yields

$$L_{n,\rho}^{\text{sym}}(\lambda) = \min\{0, -1 + c\delta_n, -1 + c\}, \quad D_{n,\rho}^{\text{sym}} = \min\{0, -1 + \rho\delta_n\}. \quad (14)$$

For the first formula, the $q = 0$ branch has value $\min(0, c)$ and is dominated by one of the three displayed terms. The second follows because $c \leq \rho$ and $\lambda = 0$ attains the bound. Thus the least exact penalty is δ_n^{-1} , and gap at most $\epsilon < 1$ requires exactly $\rho \geq (1 - \epsilon)\delta_n^{-1}$. Any strictly larger penalty with $\lambda = 0$ is minimizer-set exact. Strict points again use $a_i = 3/8$, with $y = 0$ for $q = 0$ and $y = (2b - 1)/2$ for $q = 1$; their minimum continuous inequality slack is $1/8$.

The fixed norm is essential. Let $p_n = 2^{-n}$, a rational exponent with $O(n)$ bits. Then $\delta_n^{p_n} = 1/2$. For either construction, $f + 2|y|^{p_n}$ has minimum zero: at $q = 0$ it is nonnegative and at $q = 1$ it is at least $-1 + 2\delta_n^{p_n} = 0$. Any coefficient larger than two gives minimizer-set exactness. This instance-dependent augmentation is not Lipschitz at zero and is not a norm. It changes the encoding question and the continuous minimization problem. Likewise, succinct exponent expressions or a rescaled linking equality change the representation or formulation to which Theorem 3.1 applies.

4 An encoding upper bound in continuous dimension

We now consider rational quadratic input of total binary length $N \geq 2$:

$$v = \min\{f(x, z) : (x, z) \in X, Ax + Bz - b = 0\}, \quad x \in \mathbb{R}^n, z \in \mathbb{Z}^p, \quad (15)$$

where $n \geq 1$ and the residual $r(x, z) = Ax + Bz - b$ has $m \geq 1$ coordinates. The native set is specified by finite rational coordinate bounds and inequalities $g_i(x, z) \leq 0$, all of total degree at most two; f has the same degree bound. Let s count all native inequalities, including the boxes. Affine equalities may be represented by paired inequalities. The input lists expanded polynomials, with coefficients and variable indices encoded explicitly; in particular, $s, m, n \leq N$.

For every integer assignment, assume that $f(\cdot, z)$ and each $g_i(\cdot, z)$ are convex on $\mathbb{R}^n$. Write $X_z = \{x : (x, z) \in X\}$. A slice is *equality-feasible* if $X_z \cap \{x : Ax + Bz = b\} \neq \emptyset$. Suppose the original feasible set is nonempty and, on every such slice, this intersection contains a point strictly satisfying every native inequality nonlinear in x ; native affine inequalities may hold weakly. This is the *refined Slater condition*. No margin is given, and no Slater condition is imposed on equality-infeasible slices. Empty native slices may be discarded. The finite boxes leave only finitely many integer assignments.

Write $\text{bit}(a) = 1 + \lfloor \log_2 a \rfloor$ for a positive integer a . All constants implicit below are absolute and independent of the integer dimension and the number of slices.

Theorem 4.1. *Under these assumptions there is an integer $\rho \geq 1$ with*

$$\text{bit}(\rho) \leq (N + 1)2^{O(n)} \quad (16)$$

such that $(\rho, 0)$ is minimizer-set exact for either $\|\cdot\|_\infty$ or $\|\cdot\|_1$.

We use the following consequence of effective bounds for real algebraic sets. A *basic closed set* is a conjunction of polynomial equalities and weak inequalities.

Lemma 4.2 (Effective radii). *Let at most S integer polynomials in d variables have degree at most two and coefficient bitsize at most $\tau \geq 1$. There is a radius $R \geq 1$ with*

$$\log_2 R \leq (\tau + \log(S + 1) + 1)2^{O(d)} \quad (17)$$

that meets every nonempty basic closed set defined by these polynomials and contains every such set that is bounded. The same statements hold for a finite union of such sets.

Proof. Apply the component-meeting and bounded-component radii of Basu and Roy [4, Theorems 3–4] to weak sign conditions and take the larger radius. Their explicit formulas give (17) for fixed degree. A conjunction that leaves some polynomial signs unspecified is a finite union of compatible weak sign conditions. For a union, select one nonempty member to obtain a point; if the union is bounded, every member and every component is bounded. The source proof’s factorial-bit term, retained when majorizing its meeting radius, is also absorbed by $2^{O(d)}$. $\square$

Only closed conjunctions and their finite unions are used here; the lemma makes no claim about arbitrary strict sign conditions.

4.1 Uniform coefficient bounds

The explicit input gives coordinate bounds of magnitude $2^{O(N)}$ and $O(N)$ encoded monomials. Multiplying by the product of all input denominators clears coefficients with $O(N)$ bits. Substituting any boxed integer assignment into a degree-two monomial adds only $O(N)$ bits; summing monomials adds $O(\log N)$ more. Differentiation and introducing multiplier variables preserve this uniform $O(N)$ coefficient-height bound. Thus every degree-two polynomial system below can be cleared with

$$\tau = O(N). \quad (18)$$

We clear the polynomial equations of certificate systems, not the residual in the penalty; the multiplier coordinates keep their original meaning. A box bound $|f| \leq F_0$, with an integer $F_0 \geq 1$ and $\text{bit}(F_0) = O(N)$, is equally immediate. For example, if $H \geq 1$ bounds all coordinate magnitudes and C is the sum of the absolute objective coefficients, take $F_0 = 1 + \lceil CH^2 \rceil$.

4.2 Feasible slices: a sparse multiplier certificate

Fix an equality-feasible assignment and suppress z . Refined Slater ensures a convex KKT certificate (x^*, μ, λ) , with $\mu \geq 0$. To apply the usual relative-interior form of Slater, one may blend the strictly feasible point with a point in the relative interior of the polyhedron defined by the affine constraints. For a sufficiently small blend, all nonlinear inequalities remain strict. The resulting convex optimality conditions express the normal of that polyhedron by its active affine rows, giving the stated KKT certificate even when these rows are dependent.

Let $e = \text{rank } A$ and select independent rows J spanning its row space. Project stationarity onto the quotient by this row space. The projected vector $-\nabla f(x^*)$ is in the cone generated by active native gradients, in dimension $n - e$. Conic Carathéodory therefore selects at most $n - e$ gradients, indexed by I . The remaining vector is in the row space of A . Hence $|I| + |J| \leq n$, and the following

system has a solution:

$$\begin{aligned}
g_i(x) &\leq 0 \quad (1 \leq i \leq s), & Ax + Bz - b &= 0, \\
\mu_i &\geq 0, & g_i(x) &= 0 \quad (i \in I), \\
\nabla f(x) + \sum_{i \in I} \mu_i \nabla g_i(x) + \sum_{j \in J} \lambda_j A_j^T &= 0.
\end{aligned} \tag{19}$$

This is a basic closed set in at most $2n$ variables, defined by at most $s + m + 2n$ degree-two polynomials: the active equations reuse the native polynomials g_i . Replacing complementarity products by these equations keeps degree two. The selected original certificate proves nonemptiness; every point in the new system remains a KKT certificate.

By Lemma 4.2, some certificate has all coordinates bounded by R_M , where

$$\log_2 R_M \leq (\tau + \log(s + m + 2n + 1) + 1)2^{O(n)}. \tag{20}$$

Set omitted linking multipliers to zero. The resulting multiplier has $\|\lambda_z\|_1 \leq nR_M$. Convexity, stationarity and complementarity imply, throughout the native slice,

$$f(x, z) + \lambda_z^T (Ax + Bz - b) \geq v_z. \tag{21}$$

Indeed, the full convex Lagrangian is minimized at the KKT point and has value v_z there; deleting its nonpositive native-inequality terms on X_z only increases it. We bound one suitable certificate, not all optimal multipliers, whose set can be unbounded.

4.3 Infeasible slices: a reciprocal graph

For a nonempty equality-infeasible native slice, compactness gives $\delta_z = \min_{x \in X_z} \|Ax + Bz - b\|_\infty > 0$. Write $r_j = r_j(x, z)$ and consider

$$T_z = \left\{ (x, t) : \begin{array}{l} x \in X_z, \quad t \geq 0, \quad -1 \leq tr_j \leq 1 \quad (1 \leq j \leq m), \\ tr_j = 1 \text{ or } tr_j = -1 \text{ for at least one } j \end{array} \right\}. \tag{22}$$

This is exactly the graph of $t = 1/\|r(x, z)\|_\infty$. It is nonempty and compact, since the denominator is bounded away from zero on X_z . Selecting a coordinate and a sign in the last line expresses it as a union of $2m$ basic closed sets. The equality uses one of the polynomials already present in $-1 \leq tr_j \leq 1$, so only $s + 2m + 1$ degree-two polynomials in $n + 1$ variables are needed. The bounded-set part of Lemma 4.2 gives

$$\delta_z^{-1} \leq R_D, \quad \log_2 R_D \leq (\tau + \log(s + 2m + 2) + 1)2^{O(n)}. \tag{23}$$

No KKT condition or rationality of δ_z is used on these slices.

Proof of Theorem 4.1. Choose an integer $E \geq 0$ with $2^E \geq \max\{R_M, R_D\}$ satisfying the preceding bounds, and set

$$\rho = (n + 2F_0 + 1)2^E. \tag{24}$$

This is strictly larger than both nR_M and $2F_0R_D$. Lemma 2.4, with objective bounds $-F_0, F_0$, proves minimizer-set exactness for the infinity norm. In detail, on a feasible slice a point of nonzero residual has penalized value at least $v_z + (\rho - nR_M)\|r\|_\infty > v$, while on an infeasible slice it has value at least $-F_0 + \rho/R_D > F_0 \geq v$. The same coefficient works for the one-norm because $\|r\|_1 \geq \|r\|_\infty$ and both vanish on the original feasible set. Equations (18)–(23) and $\text{bit}(F_0) = O(N)$ give (16). All bounds are uniform over boxed integer assignments, so no factor counting those assignments appears. $\square$

Together with Theorem 3.1, this gives exponential upper and lower dependence of penalty bit length on continuous dimension. The constants in the exponents and the dependence on the full sparse input length are not asserted to be sharp.

5 Polynomial encoding with few nonlinear quadratic inequalities

Let k bound the number of native inequalities nonlinear in the continuous variables after any integer assignment is fixed. Affine constraints and the quadratic objective are not counted in k .

Theorem 5.1. *Under the assumptions of Section 4, there is an integer $\rho \geq 1$ with*

$$\text{bit}(\rho) \leq N^{O(k+1)} \quad (25)$$

that is minimizer-set exact at multiplier zero for either the infinity norm or the one-norm. The exponent has an absolute constant, independent of the numbers of continuous variables, affine rows and integer assignments.

The proof allows singular quadratic Hessians and degenerate affine geometry. Its main step is an optimal-value bound without a constraint qualification. Few-quadratic algebraic methods have direct precedents: Grigoriev and Pasechnik [14, Theorem 1.2] give sampling and coefficient bounds for quadratic maps, while Kamminga and Rudolph [18, Sections 6.3–7.2 and 8.5] give reduced-variable optimization formulas for a bounded solution set and an explicit application to QCQP with a fixed total number of constraints under that boundedness hypothesis. Our convex affine-face reduction below allows arbitrary affine rows. The value bound is a consequence of established algebraic tools; the result here is its exact-penalty application and the explicit dependence on the nonlinear count.

5.1 A nonzero optimal-value bound

Lemma 5.2. *Minimize a rational convex quadratic objective over a nonempty set described by a rational polyhedron with finite coordinate bounds and at most k convex quadratic inequalities. Let $L \geq 2$ be its explicit input length and θ its optimum. Then θ is a root of a nonzero integer polynomial of degree and coefficient bitsize $L^{O(k+1)}$. In particular, if $\theta \neq 0$, then*

$$|\theta| \geq 2^{-L^{O(k+1)}}. \quad (26)$$

No Slater condition is required.

Proof. We first remove affine inequalities without changing the optimum, then eliminate the primal variables from a regularized KKT system.

Affine-face reduction. Let x^* be an optimum and let H be the affine space defined by all original equalities and all affine inequalities active at x^* . The remaining affine rows have positive slack in a neighborhood of x^* relative to H . Deleting these rows leaves x^* globally optimal over H and the nonlinear inequalities: a point of lower objective value would, by convexity, give lower values along its segment to x^* , including in that neighborhood, contradicting original optimality.

Choose independent equations defining H . Rational elimination gives $x = x_0 + Vu$ with polynomial coefficient bitsize in L . The free coordinates $u \in \mathbb{R}^d$ can be chosen among the coordinates of x . If $d = 0$, the objective value is rational of polynomial bit length and the assertion follows. Otherwise let an integer $B \geq 1$ of polynomial bit length bound the original coordinates. Then $\|u^*\|_2^2 \leq dB^2$. Add the ball $\|u\|_2^2 \leq dB^2 + 1$. It is strict at u^* , retains the same optimum, and makes the new feasible set compact. We now have an objective q_0 and $h \leq k + 1$ convex quadratic

inequalities $q_i \leq 0$, with no remaining affine constraints and coefficient bitsize polynomial in L . Some retained polynomials may become affine or constant after substitution, which does not affect the argument.

Regularization and elimination. For $0 < \varepsilon < 1$, define

$$\theta(\varepsilon) = \min\{q_0(u) + \varepsilon\|u\|_2^2 : q_i(u) \leq \varepsilon \ (1 \leq i \leq h)\}. \quad (27)$$

The point u^* strictly satisfies the relaxed inequalities. The relaxed ball gives a common bound for all feasible points. The objective is strictly convex, so an optimum and nonnegative KKT multipliers exist. Write $q_i(u) = \frac{1}{2}u^T Q_i u + a_i^T u + c_i$ with $Q_i \succeq 0$, including $i = 0$. Stationarity becomes

$$Mu = -a_0 - \sum_{i=1}^h \lambda_i a_i, \quad M = Q_0 + 2\varepsilon I + \sum_{i=1}^h \lambda_i Q_i \succ 0. \quad (28)$$

Let $\Delta = \det M$ and $p = -\text{adj}(M)(a_0 + \sum_i \lambda_i a_i)$. These are polynomials of degree at most d in (ε, λ) , and $u = p/\Delta$. Define

$$G_i = \frac{1}{2}p^T Q_i p + \Delta a_i^T p + (c_i - \varepsilon)\Delta^2 \quad (1 \leq i \leq h), \quad (29)$$

$$G_0 = \frac{1}{2}p^T Q_0 p + \Delta a_0^T p + (c_0 - w)\Delta^2 + \varepsilon p^T p. \quad (30)$$

For $0 < \varepsilon < 1$, the polynomial system

$$\mathcal{K}(\varepsilon, w, \lambda) : \quad \Delta > 0, \quad \lambda_i \geq 0, \quad G_i \leq 0, \quad \lambda_i G_i = 0 \ (1 \leq i \leq h), \quad G_0 = 0 \quad (31)$$

has a solution in λ exactly when $w = \theta(\varepsilon)$. Necessity follows from KKT. Conversely, p/Δ satisfies stationarity, feasibility and complementarity, since $G_i = \Delta^2(q_i(p/\Delta) - \varepsilon)$ and $G_0 = \Delta^2(q_0(p/\Delta) + \varepsilon\|p/\Delta\|_2^2 - w)$. Convexity then certifies global optimality.

Every polynomial in (31) has degree at most $2d + 2$. After a common denominator clears the quadratic coefficients, a determinant coefficient sums at most $d!(h+2)^d$ products of at most d input coefficients. Its bitsize is therefore polynomial in L ; the adjugate and the G_i have the same property. This bounds coefficient heights even when the number of monomials is large. Clearing the final positive denominators does not change the conditions.

The limit and its algebraic height. We have $\theta(\varepsilon) \rightarrow \theta$ as $\varepsilon \downarrow 0$. The original optimum supplies the upper limit. For the lower limit, take a convergent subsequence of regularized minimizers, using the common ball bound. Its limit satisfies all original inequalities, and continuity gives objective at least θ . No boundedness of the regularized multipliers is needed.

Consequently the following formula, with free scalar a , defines exactly $\{\theta\}$:

$$\forall \eta \left[\eta \leq 0 \vee \exists \varepsilon, w, \lambda \left(\begin{array}{l} 0 < \varepsilon < 1, \quad \varepsilon < \eta, \\ -\eta < w - a < \eta, \quad \mathcal{K}(\varepsilon, w, \lambda) \end{array} \right) \right]. \quad (32)$$

For $a = \theta$, convergence supplies a witness for every positive η . Conversely, witnesses with $\eta \downarrow 0$ give $\varepsilon \downarrow 0$ and $w \rightarrow a$, so $a = \theta$. There are two quantifier blocks of sizes 1 and $h+2$, one free variable, degree $O(d+1)$ and polynomial input coefficient bitsize. The existential block can pass the disjunct $\eta \leq 0$, whose variables it does not contain.

Effective block quantifier elimination [5, Theorem 14.16], also stated in Basu [3, Theorem 2.27], yields an equivalent quantifier-free formula with degrees and integer coefficient bitsizes $L^{O(k+1)}$. Specifically, its degree bound is $D^{O(b_1)O(b_2)}$ and its coefficient bitsize bound is $\tau D^{O(b_1)O(b_2)O(\ell)}$ for two blocks of sizes b_1, b_2 and ℓ free variables. Here $b_1 = \ell = 1$, $b_2 = h+2 \leq k+3$, $D = O(d+1)$

and $\tau = L^{O(1)}$. This uses the coefficient-height conclusion, not just the arithmetic operation bound of quantifier elimination.

Remove identically zero polynomials from that formula. At least one remaining polynomial vanishes at θ ; otherwise all its signs would be constant in a neighborhood of θ , whereas it defines a singleton. This proves the algebraic degree and height claim. If $\theta \neq 0$, remove any factor that is a power of the indeterminate from that polynomial. A Cauchy bound on the reciprocal polynomial gives $|\theta| \geq 1/(1 + 2^T)$ when T bounds its coefficient bitsize. Absorbing the additive constant in the exponent proves (26). $\square$

The argument does not identify the active affine face algorithmically, nor does it need a genericity assumption. After the affine-face reduction, squared slack variables would also convert the inequalities into a bounded quadratic-map zero set. This explains the relation to the few-constraint algebraic results cited above; the proof here instead uses a direct convex KKT elimination and an explicit two-block limit formula.

5.2 Residual separation and multiplier bounds

Proof of Theorem 5.1. Retain the objective bound $|f| \leq F_0$ from Section 4. For a nonempty equality-infeasible slice, minimize t subject to its native constraints and

$$-t \leq r_j(x, z) \leq t \quad (1 \leq j \leq m), \quad 0 \leq t \leq R_0,$$

where a rational R_0 bounds all residual magnitudes on the input box. This is a nonempty compact convex quadratic problem with only k nonlinear inequalities and positive optimum $\delta_z = \min_{X_z} \|r\|_\infty$. Its explicit length is polynomial in N , uniformly in z . Lemma 5.2 therefore gives

$$\delta_z \geq 2^{-N^{O(k+1)}}. \quad (33)$$

For an equality-feasible slice with at least one nonlinear row, let P_z be the polyhedron of native affine constraints and linking equalities. Define the common nonlinear slack

$$\sigma_z = \max\{t : x \in P_z, \ g_i(x, z) + t \leq 0 \text{ for all nonlinear rows } i, \ 0 \leq t \leq 1\}. \quad (34)$$

Refined Slater gives $\sigma_z > 0$. Minimizing $-t$ in this compact problem and applying Lemma 5.2 yields

$$\sigma_z \geq 2^{-N^{O(k+1)}}. \quad (35)$$

Let $\bar{x}$ attain this margin and take a KKT certificate at a primal slice optimum x^* . Write μ for its nonlinear multipliers. Evaluate the globally minimized convex Lagrangian at $\bar{x}$. The linking terms vanish and the native affine terms are nonpositive, so

$$v_z \leq f(\bar{x}, z) - \sigma_z \sum_i \mu_i, \quad \sum_i \mu_i \leq \frac{2F_0}{\sigma_z}. \quad (36)$$

It remains to bound linking multipliers without bounding the affine multipliers in this particular certificate. The vector

$$b_* = -\nabla f(x^*, z) - \sum_i \mu_i \nabla g_i(x^*, z)$$

has coordinates bounded by $2^{N^{O(1)}}(1 + 2F_0/\sigma_z)$, since all gradients are affine and x^*, z lie in their boxes. It belongs to the cone generated by active native affine normals and both signs of the linking

normals. Choose a conic representation with linearly independent generators, at most n in number. Restrict their column matrix to a nonsingular square row minor. Its entries are rational input coefficients after integer substitution, so determinant bounds and Cramer's rule bound its inverse entries by $2^{N^{O(1)}}$. Applying that inverse to the corresponding coordinates of b_* bounds all selected conic coefficients by $2^{N^{O(k+1)}}$. This is a bound for real coefficients; the original primal point and nonlinear multipliers need not be rational. Combining the coefficients of oppositely signed linking normals gives a new certificate with

$$\|\lambda_z\|_1 \leq 2^{N^{O(k+1)}}. \quad (37)$$

Only active affine rows were used and the nonlinear multipliers were retained, so stationarity and complementarity still hold. If there are no nonlinear rows, the same conic argument applies directly with $b_* = -\nabla f(x^*, z)$; no slack problem is needed.

All auxiliary encodings have length polynomial in N , and all constants are uniform over integer assignments. Choose an integer strictly larger than the bound in (37) and than $2F_0$ times the reciprocal bound in (33). Its bit length is $N^{O(k+1)}$. Lemma 2.4 proves minimizer-set exactness for the infinity norm, and hence for the one-norm as before. $\square$

The growing number of nonlinear inequalities in the chain construction is therefore necessary for superpolynomial penalty encoding within this model. Arbitrarily many affine rows alone do not reproduce that effect.

5.3 A conservative coefficient can be printed in polynomial time

The uniformity of the encoding bounds has an algorithmic consequence that does not require finding an optimum, a Slater point or an active face. The linear-constraint case already has polynomial encoding [15, Theorem 11] and polynomial-time sufficient-penalty output for MILPs [19, Theorem 15]; the latter source also notes the convex-MIQP extension. Here the bounds extend conservative output to native quadratic constraints when either a positive count k or the continuous dimension n is fixed.

Corollary 5.3. *For each fixed continuous dimension n , and separately for each fixed bound k on the nonlinear native count, there is a polynomial-time algorithm that, on input promised to satisfy Section 4, outputs an ordinary binary integer ρ that is minimizer-set exact at multiplier zero for both stated norms.*

Proof. Choose a universal integer constant C large enough that the bounds in Theorems 4.1 and 5.1 are at most

$$B_n = (N + 1)2^{C(n+1)}, \quad B_k = N^{C(k+1)},$$

respectively. In each case some sufficient integer has at most B_n or B_k bits and is therefore smaller than 2^{B_n} or 2^{B_k} . Increasing an exact zero-multiplier penalty preserves value exactness and excludes every nonzero-residual minimizer. Thus the algorithm may simply print a one followed by B_n or B_k zeros. For fixed n or fixed k , respectively, this output and its production time are polynomial in N . $\square$

The absolute constants exist by the cited effective bounds and may be hardcoded; the corollary does not require them as part of the input. The cited quantifier-elimination bounds give asymptotic constants, which we have not evaluated numerically; we have not implemented this conservative rule. The algorithm does not verify the input promises, solve the original mixed-integer problem, or approximate the smallest penalty. A polynomial bit bound still permits an exponentially large numerical coefficient.

6 Hardness of calibrating the smallest penalty

A short sufficient coefficient can be easy to obtain while a nearly smallest one is hard to find. We prove this for a binary box, a linear objective and one equality. The original problem has a supplied unique optimizer. Throughout this section, the least penalty means ρ_* from Section 2, with the multiplier optimized and exactness defined by dual values.

6.1 A binary-box construction

Take a SUBSET SUM instance with positive integer items $a_1, \dots, a_n$ and positive target B . Append $a_{n+1} = B + 1$ and, for an integer $K \geq 2$, set

$$X = \{0, 1\}^{n+2}, \quad f(x, q) = -q, \quad r(x, q) = K \sum_{i=1}^{n+1} a_i x_i - (KB + 1)q. \quad (38)$$

The equality $r = 0$ has the unique solution $(x, q) = (0, 0)$: at $q = 1$ it contradicts divisibility by K , and at $q = 0$ positivity forces all items to be absent. Thus the primal value is zero without solving the subset-sum instance. Linear optimization over X is also immediate; this does not make the augmented absolute-residual subproblem easy.

Let S be the attainable subset sums after appending the item, and define

$$S_- = \max\{s \in S : s \leq B\}, \quad d_- = K(B - S_-) + 1, \quad d_+ = K - 1. \quad (39)$$

The empty subset ensures that S_- exists. The smallest attainable sum above B is $B + 1$, and appending that positive item does not change whether B is attainable.

Proposition 6.1. *For (38) and every $\rho \geq 0$,*

$$D_\rho = \min \left\{ 0, -1 + \rho \frac{2d_- d_+}{d_- + d_+} \right\}, \quad \rho_* = \frac{1}{2} \left(\frac{1}{d_-} + \frac{1}{d_+} \right). \quad (40)$$

An attaining multiplier is $\lambda_\rho = \rho(d_- - d_+)/(d_- + d_+)$.

Proof. The nearest negative and positive residuals at $q = 1$ are $-d_-$ and d_+ , both with cost -1 . Their augmented values are $-1 + (\rho - \lambda)d_-$ and $-1 + (\rho + \lambda)d_+$. Weights $d_+/(d_- + d_+)$ and $d_-/(d_- + d_+)$ cancel the multiplier and give the nonzero branch in (40) as an upper bound. The feasible origin also bounds D_ρ by zero. At λ_ρ the two values agree, and $\rho \pm \lambda_\rho \geq 0$. Every other $q = 1$ residual has at least the corresponding magnitude, while every $q = 0$ residual is nonnegative and has cost zero. Thus the upper bound is attained. $\square$

At ρ_* the two infeasible witnesses tie with the primal optimum. Every strictly larger coefficient, with the multiplier fixed at λ_{ρ_*} , is minimizer-set exact. A universally sufficient value coefficient is already available: $\rho = 1$ at multiplier zero. All nonzero residuals are integers and the objective is 0 or -1 . Infeasible points can tie at coefficient one; every coefficient greater than one is minimizer-set exact at multiplier zero.

If the subset-sum instance is a YES instance, then $d_- = 1$; if it is NO, then $d_- \geq K + 1$. Consequently

$$\rho_* = \frac{K}{2(K-1)} > \frac{1}{2} \quad \text{in YES instances,} \quad 0 < \rho_* \leq \frac{K}{K^2-1} \quad \text{in NO instances.} \quad (41)$$

Theorem 6.2. *On the family (38) with $K = 4$, deciding whether $D_{1/3} = 0$ is coNP-complete. More precisely, $D_{1/3} = -1/2$ in YES subset-sum instances and $D_{1/3} = 0$ in NO instances.*

Proof. In YES instances, $d_- = 1$, $d_+ = 3$, and (40) gives $D_{1/3} = -1/2$. In NO instances, $\rho_* \leq 4/15 < 1/3$, so $D_{1/3} = 0$. This is a polynomial reduction from the complement of SUBSET SUM. On this specified family, a subset summing to B is a polynomial certificate of nonexactness, proving membership in coNP. $\square$

Thus computing ρ_* is NP-hard, and evaluating $D_{1/3}$ to absolute error strictly below $1/4$ would decide SUBSET SUM. By contrast, $D_0 = -1$ for every instance of this family.

Theorem 6.3. *Fix any polynomial p with $p(L) \geq 1$. Unless $P = NP$, no polynomial-time algorithm returns, on every instance (38) of binary length L , a rational $\hat{\rho}$ satisfying*

$$\rho_* \leq \hat{\rho} \leq p(L)\rho_*. \quad (42)$$

Proof. Let ℓ be the input length of the original subset-sum instance and choose $K = 2^\ell$. There are $O(\ell)$ coefficients, each of $O(\ell)$ bits, so the constructed length is $L = O(\ell^2)$ and the construction takes polynomial time. Every admissible YES output exceeds $1/2$. Every admissible NO output is at most

$$p(L) \frac{2^\ell}{2^{2\ell} - 1} < \frac{1}{2}$$

for all sufficiently large ℓ , since an exponential dominates the fixed polynomial $p(O(\ell^2))$. The finitely many smaller inputs can be handled by exhaustive enumeration. Comparing the output with $1/2$ would therefore decide SUBSET SUM in polynomial time. $\square$

The same construction rules out a symmetric estimate $\rho_*/p(L) \leq \hat{\rho} \leq p(L)\rho_*$: YES outputs exceed $1/(2p(L))$ and NO outputs are smaller for sufficiently large ℓ . For a fixed one-sided factor C , a constant K with $K + 1 > 2C$ suffices, because the ratio between the YES threshold and the NO upper bound is $(K + 1)/2$. Positive rational scaling of the equality rescales both penalty thresholds by the reciprocal factor and preserves this relative gap. It does not remove the small arithmetic spacing. The binary-box result is a bit-complexity obstruction, not strong hardness or an APX-completeness classification.

The same calibration obstruction holds with the multiplier fixed at zero. The two nearest improving residuals give

$$\rho_*(0) = \max\{1/d_-, 1/d_+\}.$$

Indeed, all other improving points have larger residual magnitude, and the cost-zero points impose no positive requirement. For $K \geq 3$ this threshold is one in YES instances and $1/(K - 1)$ in NO instances. Taking $K = 2^\ell$ in the proof of Theorem 6.3 therefore also excludes polynomial-factor sufficient calibration at multiplier zero; the finitely many small inputs can again be enumerated.

6.2 Unit coefficients with a harder native set

For a nonempty simple graph $G = (V, E)$, take binary variables x_v, p_+, p_- with native constraints

$$p_+ + p_- \leq 1, \quad x_v \leq p_+ + p_- \quad (v \in V), \quad x_u + x_v \leq 1 \quad (uv \in E). \quad (43)$$

Set $f = -\sum_v x_v$ and $r = p_+ - p_-$. The equality again forces the unique primal point zero. If $\alpha(G)$ is the maximum stable-set size, the native branches have residual zero and cost zero, or residual ± 1 and minimum cost $-\alpha(G)$. Hence

$$L_\rho(\lambda) = \min\{0, \rho - \alpha(G) - |\lambda|\}, \quad D_\rho = \min\{0, \rho - \alpha(G)\}, \quad \rho_* = \alpha(G). \quad (44)$$

Computing ρ_* is strongly NP-hard with all model coefficients and right-hand sides in $\{-1, 0, 1\}$. Exactness at an integer k is the coNP-complete condition $\alpha(G) \leq k$; a stable set of size $k+1$ certifies failure. At equality, infeasible maximum stable sets tie. At multiplier zero, minimizer-set exactness requires $\rho > \alpha(G)$. Here the native optimization is already hard, unlike linear optimization over the box in (38).

6.3 Relation to penalty-choice hardness

The growing-binary-dimension results answer the smallest-penalty question in the conclusion of Lefebvre and Schmidt [19]; they do not assert hardness in fixed total dimension. Conservative sufficient bounds are also established: for binary quadratic problems with integer linear constraints, Gusmeroli and Wiegele [17, Sections 5–6 of the inspected preprint] obtain exact squared penalties from objective bounds and residual separation. Our statement concerns approximation of the instance’s least norm-penalty threshold, not the general idea of using a safe upper bound. Generic penalty-choice hardness, even with a known constrained optimum, is older. Alessandrini et al. [1, Section IV.A, Lemma 1 of the inspected preprint] prove hardness for exact QUBO reformulations. Their reduction imposes $x = 0$ and uses a squared vector residual and a quadratic objective. On binary variables, its penalty is $M \sum_i x_i$, which unrestricted linear multipliers reproduce with augmentation coefficient zero. Thus that reduction does not give a positive optimized-dual threshold. Our restrictions and polynomial-factor conclusion are the distinction.

Other tuning results use different quantifiers. Mirkarimi et al. [20, Section 2.2] note that a universally successful, polynomially tuned linear Ising penalty for knapsack would imply $P = NP$, allowing the possibility that no suitable linear penalty exists on some instances. Alessandrini et al. [2, Definition 1 and Theorem 2] instead study probabilistic feasibility and objective guarantees when exact Gibbs sampling is used for approximate optimization. These conclusions differ from approximating an always-existing positive norm-penalty threshold after unrestricted multiplier optimization.

7 Right-hand-side perturbations

The worst-case bounds concern a fixed equality. We now perturb its right-hand side and prove a finite-grid encoding guarantee. This changes the optimization problem; exactness is asserted for the sampled instance. The conditioning principle is classical: avoiding convex boundaries controls distance to feasibility changes [11]; spherical uniform-noise boundary estimates are also known [10, Theorem 3.3 and Corollary 3.4 of the inspected preprint]. The contribution in this section is the explicit finite-grid exact-penalty guarantee, including its bit length and its feasibility alternative. We prove the cube estimate and grid transfer used for this application; the Gaussian result later uses Dunagan et al. [11, Lemma 2.3.4].

There are at most $K \geq 1$ nonempty compact convex native slices X_z . On each slice, $r_z : X_z \rightarrow \mathbb{R}^m$ is affine and f_z is continuous and convex, with supplied common bounds $L \leq f_z \leq U$ and range $M = U - L$. All slices, maps and objectives are fixed before sampling. For a right-hand side b , let

$$\begin{aligned} v(b) &= \min\{f_z(x) : z \in Z, x \in X_z, r_z(x) = b\}, \\ p_\rho(b) &= \min_{z \in Z, x \in X_z} \{f_z(x) + \rho \|r_z(x) - b\|_\infty\}, \end{aligned} \tag{45}$$

where $v(b) = +\infty$ if infeasible. At a feasible b , $\rho_*(b)$ denotes the optimized threshold of Section 2 for residual $r_z(x) - b$ and the infinity norm. Each image $C_z = r_z(X_z)$ is compact and convex.

Boundaries are ambient topological boundaries; a lower-dimensional closed convex set is its own boundary. Distance to the empty set is $+\infty$.

7.1 A deterministic boundary-margin bound

Lemma 7.1. *Suppose $v(b) < \infty$, $d > 0$, and $\text{dist}_\infty(b, \partial C_z) > d$ for every slice. Then $p_{M/d}(b) = v(b)$. Every strictly larger coefficient is minimizer-set exact at multiplier zero.*

Proof. On a feasible slice, the cube $b + [-d, d]^m$ is contained in C_z : otherwise a segment from b to a point outside the image would cross its boundary within distance d . For a native point x with $t = \|r_z(x) - b\|_\infty > 0$, choose $x' \in X_z$ with

$$r_z(x') = b - \frac{d}{t}(r_z(x) - b).$$

Then $x_b = (dx + tx')/(d + t)$ is feasible at b . Convexity gives

$$v_z(b) \leq \frac{d}{d+t}f_z(x) + \frac{t}{d+t}U, \quad f_z(x) \geq v_z(b) - \frac{U - v_z(b)}{d}t \geq v(b) - \frac{M}{d}t.$$

The last inequality also holds at $t = 0$. On an infeasible slice, a closest image point lies on its boundary, so $t > d$ everywhere and $f_z(x) + (M/d)t \geq L + M \geq v(b)$. A primal optimum attains the lower bound $v(b)$. Increasing the coefficient strictly adds a positive amount to every nonzero-residual point. If $M = 0$, the zero coefficient gives equality of values and a positive coefficient gives the asserted minimizer-set conclusion. $\square$

Neither differentiability nor computed multipliers are required. The relation between convex exact penalties and multiplier bounds is classical; see Friedlander and Tseng [13, Theorem 4.2] for a polar-gauge characterization under a nonempty compact solution set. Convexity of the objective is essential: on $[-1, 1]$ with $r(x) = x$, $b = 0$ and $f(x) = -\sqrt{|x|}$, the image-boundary distance and objective range are both one, but the balanced pair $x = \pm t$ gives average augmented value $-\sqrt{t} + \rho t < 0$ for small $t > 0$, at every multiplier. No finite penalty closes that dual gap.

7.2 Convex-boundary tubes in a cube

Lemma 7.2. *Let $Q = b_0 + [-\sigma, \sigma]^m$, $\sigma > 0$, and let B be uniform on Q . For any closed convex $C \subseteq \mathbb{R}^m$ and $d \geq 0$,*

$$\Pr\{\text{dist}_\infty(B, \partial C) \leq d\} \leq \min\{1, 2md/\sigma\}. \quad (46)$$

The same upper bound holds with Euclidean distance in the event.

Proof. Fix $r > 0$ and $S_i = [-r, r]e_i$. Addition $A + S_i$ and erosion $A \ominus S_i = \{x : x + S_i \subseteq A\}$ preserve convexity and closedness. On a line parallel to e_i , a section of a closed convex set is an interval, possibly empty, a point, a ray or the whole line. Addition extends each finite endpoint by r . Erosion removes at most r at each finite endpoint, possibly deleting a short interval. After clipping to the corresponding section of Q , either operation changes length by at most $2r$. Thus Fubini's theorem gives

$$\begin{aligned} \text{vol}(((A + S_i) \setminus A) \cap Q) &\leq 2r(2\sigma)^{m-1}, \\ \text{vol}((A \setminus (A \ominus S_i)) \cap Q) &\leq 2r(2\sigma)^{m-1}. \end{aligned}$$

Only changed portions inside Q are measured, so no infinite volumes are subtracted. Successive additions give $A + [-r, r]^m$; successive erosions give $A \ominus [-r, r]^m$, since $(A \ominus S) \ominus T = A \ominus (S + T)$. Telescoping the nested sets bounds each resulting shell inside Q by $2mr(2\sigma)^{m-1}$.

Take $r > d$. A point within distance d of ∂C is in $C + [-r, r]^m$ and is not in $C \ominus [-r, r]^m$: a nearby boundary point has exterior points arbitrarily close to it, one of which lies in that cube about the given point. Hence the closed d -tube lies in the union of the two shells. Dividing their volume bound by $(2\sigma)^m$ and letting $r \downarrow d$ proves (46), including $d = 0$. This argument covers unbounded and lower-dimensional sets. For $C = \emptyset$ or $\mathbb{R}^m$, the boundary is empty and the claim is immediate. Finally, distance in the infinity norm is no greater than Euclidean distance, so the latter event is contained in the former. $\square$

The coefficient $2m$ is sharp for bounds of the form $c_m d/\sigma$. For $C = [-a, a]^m$, $0 < a < \sigma$, and small $d > 0$, the tube probability is $((a+d)^m - (a-d)^m)/\sigma^m$. Its derivative at zero is $(2m/\sigma)(a/\sigma)^{m-1}$; let $a \uparrow \sigma$.

7.3 Rational-grid exactness and its encoding

For an integer $q \geq 1$, sample each coordinate of G independently and uniformly from

$$b_{0i} - \sigma + (j + 1/2)h, \quad j = 0, \dots, q-1, \quad h = 2\sigma/q. \quad (47)$$

Adding independent uniform jitter from $[-h/2, h/2]^m$ makes $B = G + J$ uniform on Q , with $\|B - G\|_\infty \leq \sigma/q$. The distance function is one-Lipschitz, so Lemma 7.2 gives

$$\Pr\{\text{dist}_\infty(G, \partial C) \leq d\} \leq \min\{1, 2m(d/\sigma + 1/q)\}. \quad (48)$$

This rounding term is necessary: in dimension one, an interval whose two endpoints are distinct grid points has boundary probability $2/q$ at $d = 0$, exactly the displayed correction. A continuous null-set argument alone cannot justify a grid guarantee.

Theorem 7.3. *Let $b_0, \sigma, \epsilon, L, U$ be rational, with $\sigma > 0$ and $0 < \epsilon < 1$. If*

$$q \geq \left\lceil \frac{4mK}{\epsilon} \right\rceil, \quad d = \frac{\sigma\epsilon}{4mK}, \quad \bar{\rho} = \frac{4mKM}{\sigma\epsilon}, \quad (49)$$

then with probability at least $1 - \epsilon$ the sampled problem is infeasible or $p_{\bar{\rho}}(G) = v(G)$. On the same event, whenever it is feasible, every coefficient larger than $\bar{\rho}$ is minimizer-set exact at multiplier zero.

Proof. A union bound over the fixed images gives failure probability at most $2mK(d/\sigma + 1/q) \leq \epsilon$ for simultaneous strict boundary distance greater than d . Apply Lemma 7.1. $\square$

Choose q as the smallest power of two meeting the bound. Then $\log_2 q = O(\log m + \log K + \log(1/\epsilon))$ and exact uniform sampling uses $m \log_2 q$ independent random bits. The grid points and $\bar{\rho}$ have encoding length polynomial in the supplied rational data, m and $\log K$. An integer coefficient strictly larger than $\bar{\rho}$ has the same bound. For boxed integer variables, the product of their numbers of allowed integer values supplies K , with polynomially bounded $\log K$. Neither computing this coefficient nor sampling requires enumerating slices. No efficient recognition of the structural promises or global optimization procedure is asserted.

An endpoint grid with $q + 1$ points per coordinate, $b_{0i} - \sigma + jh$ for $0 \leq j \leq q$, has a similar bound. Its jitter cells tile the larger cube of radius $\sigma + h/2$, giving

$$\Pr\{\text{dist}_\infty(G, \partial C) \leq d\} \leq \min \left\{ 1, \frac{2m(d + h/2)}{\sigma + h/2} \right\} \leq \min\{1, 2m(d/\sigma + 1/q)\}.$$

Thus the theorem also holds for that convention, with the appropriate uniform grid sampling. For either grid, the Euclidean boundary event obeys the same estimate by inclusion and infinity-norm jitter; this observation alone does not replace the cube margin used by the penalty.

7.4 Feasibility, stability and the original instance

The infeasibility alternative cannot be dropped. For example, a finite union of lower-dimensional images is hit with probability zero under continuous full-dimensional noise. If a known slice satisfies $Q \subseteq C_{z_0}$, every centered-grid sample is feasible and the theorem gives unconditional exactness probability $1 - \epsilon$. More generally, if feasibility has probability at least $p_0 > 0$, then the conditional failure probability given feasibility is at most ϵ/p_0 . To obtain conditional failure at most ϵ , use ϵp_0 in the construction.

At a feasible point b with all image-boundary distances greater than d , the set of feasible integer assignments is unchanged throughout $b + [-d/2, d/2]^m$. Every point there retains boundary margin greater than $d/2$. Applying the repair inequality in both directions shows that each feasible slice value is $2M/d$ -Lipschitz in the infinity norm on that cube. Taking the minimum over the same finite collection of slice values preserves this Lipschitz bound. The coefficient $2M/d$ is value-exact throughout the cube, and every larger one is minimizer-set exact there. Lipschitz value bounds across discrete choices under uniform Slater conditions also appear in mixed-integer optimal control [16, Theorem 3].

This local conclusion does not recover the original optimum at b_0 . In the chain example, moving the right-hand side from zero to δ_n makes the improving binary assignment feasible and changes the optimum from zero to -1 . Shorter penalty encodings after perturbation can therefore accompany a change in the optimal integer assignment.

7.5 Numerical size: unavoidable tails and slice dependence

Let an anchor slice have residual image $[0, 1]$ and cost zero, and let $K - 1$ singleton slices have images j/K , $1 \leq j \leq K - 1$, and cost -1 . At a non-atom $b \in (0, 1)$, the primal value is zero. For a singleton at distance $t > 0$, choose an anchor point with opposite residual of magnitude $s > 0$. The balanced mixture with weights $s/(s + t)$ and $t/(s + t)$ gives augmented value

$$\frac{s}{s + t}(-1 + 2\rho t), \quad (50)$$

independently of the multiplier. Thus the optimized-dual threshold is at least $1/(2t)$. If B is uniform on $[0, 1]$, the disjoint intervals $|B - j/K| < 1/(2T)$ have total length $(K - 1)/T$ for $T \geq K$. Consequently

$$\Pr\{\rho_*(B) > T\} \geq (K - 1)/T \quad (T \geq K). \quad (51)$$

The order K/ϵ in a high-probability numerical bound is therefore necessary in this slice model, even with robust feasibility and optimized multipliers. For the centered grid on $[0, 1]$, if K divides q and $q \geq 2K$, each of the $K - 1$ singleton images has two adjacent centers at distance $1/(2q)$. These $2(K - 1)$ centers are distinct, so $\Pr\{\rho_*(G) \geq q\} \geq 2(K - 1)/q$.

For an almost-sure slice-count obstruction, put singleton images at j/J , $j = 0, \dots, J$, with cost -1 , and retain the anchor. There are $K = J + 2$ slices. Between neighboring grid points, the singleton residual magnitudes $u, v > 0$ satisfy $u + v = 1/J$. Their balanced mixture has cost -1 and mean absolute residual $2uv/(u + v) \leq 1/(2J)$. Thus $\rho_*(b) \geq 2J$ for almost every $b \in [0, 1]$. These examples concern a specified collection of slices; no succinct encoding of every such collection is assumed.

Even two slices can give an infinite mean numerical penalty. Take an anchor image $[-1, 1]$ of cost zero and a singleton at zero of cost -1 . For $b \in (-1, 1) \setminus \{0\}$, (50) and the multiplier $\lambda = -\rho \operatorname{sign}(b)$ prove

$$\rho_*(b) = \frac{1}{2|b|}. \quad (52)$$

Indeed, at that threshold the anchor's augmented objective is nonnegative and the singleton's is zero. The zero-multiplier threshold is instead $1/|b|$. Uniform continuous noise on $[-1, 1]$ gives $\mathbb{E}\rho_*(B) = +\infty$, while $\mathbb{E} \log(1 + \rho_*(B)) < \infty$ by integration near zero. The finite-grid theorem is a high-probability statement; its exceptional atoms can even require an infinite threshold. Consider the convex quadratic slice $0 \leq t \leq 1$, $t^2 \leq s \leq 1$, with cost $-t$ and image coordinate s , and a second singleton slice with image $\{-1\}$ and cost zero. At right-hand side zero the primal value is zero. Mixing $(t, s) = (u, u^2)$ with the singleton, using weights $1/(1 + u^2)$ and $u^2/(1 + u^2)$, gives cost $-u/(1 + u^2)$ and mean absolute residual $2u^2/(1 + u^2)$. Its threshold ratio is $1/(2u)$, unbounded as $u \downarrow 0$. Thus $\rho_*(0) = +\infty$. A rational grid containing zero gives this event positive probability, so no finite expected threshold or penalty-encoding requirement follows in general. The boundary-margin condition properly excludes that atom; refined Slater also fails there.

7.6 A Gaussian companion in the Euclidean norm

The deterministic repair proof works with any norm when its ball and its distance are used consistently. For Gaussian noise, use the Euclidean norm for both. Let $B = b_0 + \sigma\xi$, $\xi \sim \mathcal{N}(0, I_m)$. Dunagan et al. [11, Lemma 2.3.4] bound each of the inner and outer Euclidean tubes of a convex body by $4m^{1/4}d/\sigma$. Their sum and a union bound give

$$\Pr\{\min_z \operatorname{dist}_2(B, \partial C_z) \leq d\} \leq 8Km^{1/4}d/\sigma. \quad (53)$$

This also covers lower-dimensional compact images. To see this, replace C by its parallel body $C + \eta\mathbb{B}_2$, $\eta > 0$. For a lower-dimensional C , the distances to these boundaries converge to $\operatorname{dist}_2(\cdot, C)$ as $\eta \downarrow 0$: outside C use the distance to a parallel body, and at a point of C use a normal to its affine hull. Apply the convex-body bound at any radius greater than d , then let $\eta \downarrow 0$ and the radius decrease to d . Thus, with probability at least $1 - \epsilon$, the sampled instance is infeasible or the Euclidean zero-multiplier coefficient

$$\frac{8MKm^{1/4}}{\sigma\epsilon} \quad (54)$$

is value-exact. A larger coefficient gives minimizer-set exactness. This is a direct Gaussian conditioning corollary, with a different noise model and norm from the rational-grid encoding result.

8 Discussion

The main conclusion is a limit on extending polynomial penalty encoding from linear to convex quadratic native constraints. Finite exactness and uniform strict feasibility on feasible integer slices do not suffice: an improving infeasible slice can lie doubly exponentially close to the linking equality. The upper bounds identify two ways to recover polynomial encoding, by fixing continuous dimension or the number of nonlinear native inequalities under the stated regularity assumptions. These results refine the quantitative scope of existing exactness theorems; they do not replace their existence theory.

The calibration result then shows why a safe sufficient coefficient and a nearly smallest one are different computational targets, even when the original optimizer is supplied. The perturbation

result provides a probabilistic alternative with explicit sampling and feasibility qualifications. The encoding upper bounds and the perturbation guarantee allow large numerical coefficients and do not imply fast solution of the augmented subproblem. Changing the equality, using succinct exponents or choosing a fractional-power augmentation changes the representation question. The contributions concern ordinary binary coefficients for the prescribed norm and residual.

Supplementary scripts check the exact finite envelopes, determinant identities, degenerate regularization, calibration dual formulas and thresholds, and selected grid and tail identities with rational or symbolic arithmetic. These checks support the algebra and indexing; the proofs establish the general claims. No solver-performance claim is made.

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A An exact dual value without an attaining multiplier

Example 13 of the December 15, 2025 manuscript of Lefebvre and Schmidt [19] is stated to admit no finite exact penalty representation. Under its supremum-based Definition 1, however, the example has an exact dual value at every nonnegative penalty. In renamed variables its native set is

$$X = \{(u, s, z) \in [-1, 1]^2 \times \{0, 1\} : u + 2z \leq 1, u^2 \leq s\}, \quad f = u - z/2, \quad r = s.$$

The equality $s = 0$ forces $u = z = 0$, so $v = 0$. All native residuals are nonnegative. Proposition 2.1 therefore gives $D_\rho = 0$ for every $\rho \geq 0$.

This can also be checked directly. Put $t = \lambda + \rho$ and assume $t \geq 1/2$. The branch $z = 1$ contains only $(-1, 1, 1)$ and has value $t - 3/2$. On $z = 0$, minimization first sets $s = u^2$ and then $u = -1/(2t)$, giving

$$L_\rho(\lambda) = \min\{t - 3/2, -1/(4t)\}.$$

This tends to zero as $\lambda \rightarrow +\infty$. For any finite t , however, a sufficiently small $\eta > 0$ gives the native point $(-\eta, \eta^2, 0)$ with value $-\eta + t\eta^2 < 0$. Thus no finite multiplier attains the exact dual value. The source’s reduction to multiplier zero invokes its Lemma 3, which transfers attained exactness between fixed multipliers and does not apply to this nonattained supremum. Thus the example demonstrates nonattainment, not failure of dual-value exactness. This correction does not affect the source’s Theorem 14 under slice Slater conditions, nor the attained lower-bound formulas in Section 3.

For genuine failure of finite dual-value exactness without refined Slater, the two-sided exceptional-atom construction in Section 7 has an infinite optimized threshold at right-hand side zero.